Problem Set 4

Course Name: Modeling and Control	LP / Year: 4 / 2025	Course Code: R0004E
Number of questions: 7	**Release Date:** 2025-05-12	**Submission Date:**

1. Explain briefly, what each of the following plots are showing:
   (a) Bode plot (both the magnitude and the phase plots)
   (b) Nyquist plot

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2. Compare the following pairs of controllers/compensators, and explain the advantage of the latter with respect to earlier.
   (a) A PI controller with a lag compensator
   (b) a PD controller with a lead compensator

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3. Given a radar tracking system as shown in Figure 1.

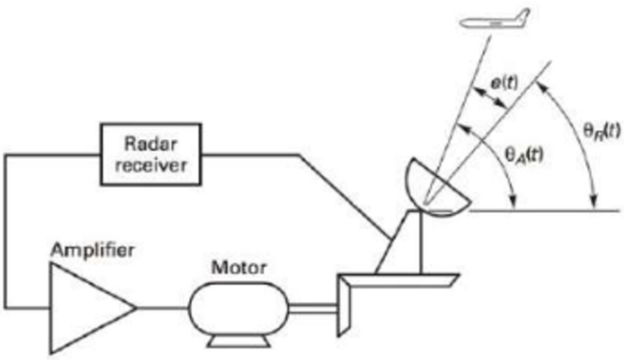

Figure 1: Radar tracking system.

The inductance of the servomotor cannot be ignored and the open-loop function of the uncompensated system is given by:

$$G_p(s)H(s) = \frac{4}{s(s+1)(s+2)}$$

with $H(s) = 1$. The required phase margin is specified to be 50^o and settling time T_s is less than 4 seconds.

Design a lead compensator, so that the specifications above are satisfied for the closed-loop system.

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4. Given a plant with the transfer function:

$$G(s) = \frac{K_g}{(s+2)(s+\alpha)}$$

   (a) Write the closed-loop transfer function of the system with a unity feedback.

   (b) Determine the value of K_g and α such that the closed-loop system satisfies all of the following criteria:

   - The steady state error for a unit step input to be less than 0.1
   - The undamped natural frequency to be greater than 15 rad/sec
   - The damping ratio to be 0.5

   (c) Having in mind the PID controller and its variants, if the damping of the closed-loop system needs to be improved, please suggest which variant should be applied to this system.

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5. Draw the Nyquist diagram for the system shown in Figure 2 and determine the value of P, N and Z, and the stability of the closed loop system.

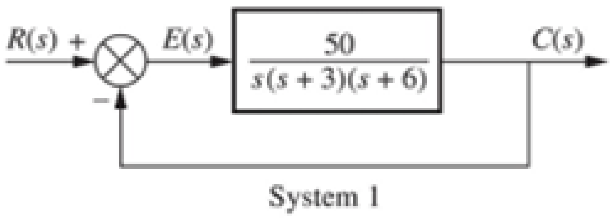

Figure 2: Nyquist plot for a system

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6. Determine the stability of the following systems, with

$$G(s) = \frac{-(s^2+4s+6)}{s^2+5s+4}$$

based on the Nyquist criteria:

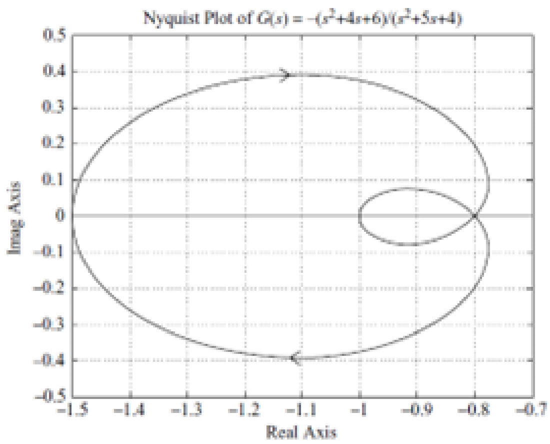

Figure 3: Nyquist plot for $G(s)$.

7. For the following system:

$$G(s) = \frac{K}{(s+2)(s+4)(s+6)}$$

(a) Find the magnitude and phase, where the Nyquist diagram of the system cross the real axis, for $K = 1$.

(b) Sketch the Nyquist diagram for $K = 1$.

(c) Find the range of the value of K for which the unity feedback system is stable.

END